Question 1:

Find the interval of convergence and radius of convergence for the power series $\sum_{n=0}^{\infty}(4x)^n$, centered at $x_0 = 0$.

Note: $\sum_{n=0}^{\infty}(4x)^n = (4x)^0 + (4x)^1 + (4x)^2 + (4x)^3 + \ldots = 1 + (4x)^1 + (4x)^2 + (4x)^3 + \ldots$

The power series $\sum_{n=0}^{\infty}(4x)^n$ is a geometric power series of the form $\sum_{n=0}^{\infty} a \cdot r^n$,

where the first term is $a = 1$ and the common ratio is $r = 4x$.

a) Find all values of x for which the power series $\sum_{n=0}^{\infty}(4x)^n$ converges.

b) When $x = -1/4$, $\sum_{n=0}^{\infty}(4x)^n = \sum_{n=0}^{\infty}(4(-1/4))^n = \sum_{n=0}^{\infty}(-1)^n$. The series $\sum_{n=0}^{\infty}(-1)^n$ converges or diverges?

c) When $x = 1/4$, $\sum_{n=0}^{\infty}(4x)^n = \sum_{n=0}^{\infty}(4(1/4))^n = \sum_{n=0}^{\infty}(1)^n$. The series $\sum_{n=0}^{\infty}(1)^n$ converges or diverges?

d) The interval of convergence for the power series $\sum_{n=0}^{\infty}(4x)^n$ is ____?____.

e) The radius of convergence for the power series $\sum_{n=0}^{\infty}(4x)^n$ is ____?____.

Question 2:

Find the interval of convergence and radius of convergence for the power series $\sum_{n=0}^{\infty}(2x-5)^n$, centered at $x_0 = 2.5$.

Note: $\sum_{n=0}^{\infty}(2x-5)^n = (2x-5)^0 + (2x-5)^1 + (2x-5)^2 + (2x-5)^3 + \ldots = 1 + (2x-5)^1 + (2x-5)^2 + (2x-5)^3 + \ldots$

The power series $\sum_{n=0}^{\infty}(2x-5)^n$ is a geometric power series of the form $\sum_{n=0}^{\infty} a \cdot r^n$,

where the first term is $a = 1$ and the common ratio is $r = 2x-5$.

a) Find all values of x for which the power series $\sum_{n=0}^{\infty}(4x)^n$ converges.

b) When $x = 2$, $\sum_{n=0}^{\infty}(2x-5)^n = \sum_{n=0}^{\infty}(2(2)-5)^n = \sum_{n=0}^{\infty}(-1)^n$. The series $\sum_{n=0}^{\infty}(-1)^n$ converges or diverges?

c) When $x = 3$, $\sum_{n=0}^{\infty}(2x-5)^n = \sum_{n=0}^{\infty}(2(3)-5)^n = \sum_{n=0}^{\infty}(1)^n$. The series $\sum_{n=0}^{\infty}(1)^n$ converges or diverges?

d) The interval of convergence for the power series $\sum_{n=0}^{\infty}(2x-5)^n$ is ____?____.

e) The radius of convergence for the power series $\sum_{n=0}^{\infty}(2x-5)^n$ is ____?____.

Question 3:

Find the power series representation for the function $f(x)=\dfrac{1}{x-3}$, centered at $x_0=0$.

Note:

Note: A geometric power series has the form $\sum_{n=0}^{\infty} a\cdot r^n$ and converges to $\dfrac{a}{1-r}$ if $-1<r<1$.

The function $f(x)=\dfrac{1}{x-3}=\dfrac{-1}{3-x}=\dfrac{-1/3}{1-\frac{1}{3}x}$ is also of the form $\dfrac{a}{1-r}$, where $a=-1/3$ and $r=\dfrac{1}{3}x$.

a) Power series representation for the function $f(x)=\dfrac{1}{x-3}$ is __?__.

b) For what values of x does the power series in part (a) converge?

c) Interval of convergence for the power series in part (a) is __?__.

d) Radius of convergence for the power series in part (a) is __?__.

e) Can the function $f(x)=\dfrac{1}{x-3}$ be expanded into a Taylor series for values near $x_0=0$?

f) Is the function $f(x)=\dfrac{1}{x-3}$ analytic at at $x_0=0$?

Question 4:

Find the power series representation for the function $f(x)=\dfrac{1}{x-3}$, centered at $x_0=2$.

Note: $f(x)=\dfrac{1}{x-3}=\dfrac{1}{(x-2+2)-3}=\dfrac{1}{(x-2)+2-3}=\dfrac{1}{(x-2)-1}=\dfrac{1}{-1+(x-2)}=\dfrac{-1}{1-(x-2)}$

Note: A geometric power series has the form $\sum_{n=0}^{\infty} a\cdot r^n$ and converges to $\dfrac{a}{1-r}$ if $-1<r<1$.

The function $f(x)=\dfrac{-1}{1-(x-2)}$ is also of the form $\dfrac{a}{1-r}$, where $a=-1$ and $r=x-2$.

a) Power series representation for the function $f(x)=\dfrac{1}{x-3}=\dfrac{-1}{1-(x-2)}$ is __?__.

b) For what values of x does the power series in part (a) converge?

c) Interval of convergence for the power series in part (a) is __?__.

d) Radius of convergence for the power series in part (a) is __?__.

e) Can the function $f(x)=\dfrac{1}{x-3}=\dfrac{-1}{1-(x-2)}$ be expanded into a Taylor series for values near $x_0=2$?

f) Is the function $f(x)=\dfrac{1}{x-3}=\dfrac{-1}{1-(x-2)}$ analytic at at $x_0=2$?

Question 5:

Find the power series representation for the function $f(x)=\dfrac{1}{x-3}$, centered at $x_0=3$.

a) $f'(x)=$ __?__ and $f'(3)=$ __?__.

b) Does a power series representation for the function $f(x)=\dfrac{1}{x-3}$ exist at $x_0=3$?

c) Can the function $f(x)=\dfrac{1}{x-3}$ be expanded into a Taylor series for values near $x_0=3$?

d) Is the function $f(x)=\dfrac{1}{x-3}$ analytic at $x_0=3$?

Question 6:

Find a power series representation for the function $f(x)=\dfrac{1}{1+x^2}$, centered at $x_0=0$.

Note: A geometric power series has the form $\sum_{n=0}^{\infty} a\cdot r^n$ and converges to $\dfrac{a}{1-r}$ if $-1<r<1$.

The function $f(x)=\dfrac{1}{1+x^2}=\dfrac{1}{1-\left(-x^2\right)}$ is also of the form $\dfrac{a}{1-r}$, where $a=1$ and $r=\left(-x^2\right)$.

a) Power series representation for the function $f(x)=f(x)=\dfrac{1}{1+x^2}=\dfrac{1}{1-\left(-x^2\right)}$ is $\underline{\quad ? \quad}$.

b) For what values of x does the power series in part (a) converge?

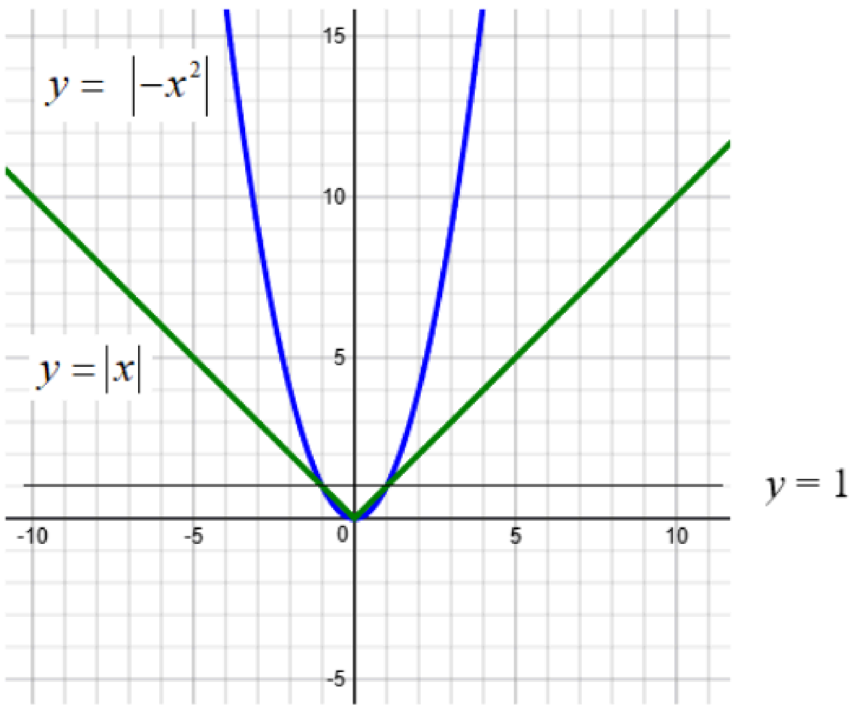
Hint: $\left|-x^2\right|<1 \quad \Rightarrow \quad x^2<1 \quad \Rightarrow \quad |x|<1$

c) Interval of convergence for the power series in part (a) is $\underline{\quad ? \quad}$.

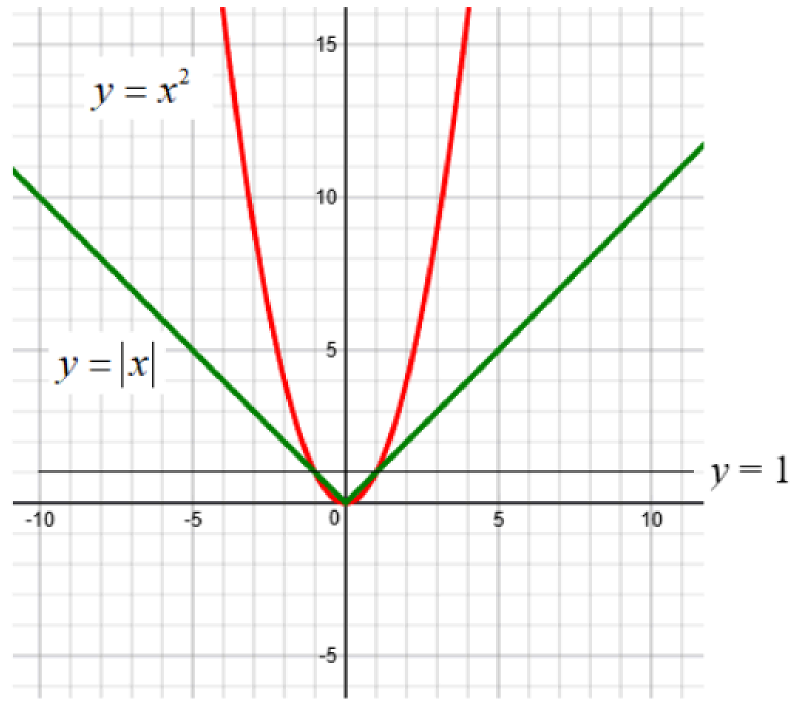
d) Radius of convergence for the power series in part (a) is $\underline{\quad ? \quad}$.

e) Can the function $f(x)=\dfrac{1}{1+x^2}$ be expanded into a Taylor series for values near $x_0=0$?

f) Is the function $f(x)=\dfrac{1}{1+x^2}$ analytic at at $x_0=0$?



$y = |-x^2|$
$y = |x|$
$y = 1$
15
10
5
-5
-10
-5
0
5
10



$y = x^2$
$y = |x|$
$y = 1$
15
10
5
-5
-10
-5
0
5
10

Question 7:

Find a power series representation for the function $f(x) = \dfrac{1}{(1-x)^2}$, centered at $x_0 = 0$.

Note: A geometric power series has the form $\sum_{n=0}^{\infty} a \cdot r^n$ and converges to $\dfrac{a}{1-r}$ if $-1 < r < 1$.

The function $f(x) = \dfrac{1}{1-x}$ is also of the form $\dfrac{a}{1-r}$, where $a = 1$ and $r = x$.

Hence, $f(x) = \dfrac{1}{1-x} = \dfrac{a}{1-r} = \sum_{n=0}^{\infty} a \cdot r^n = \sum_{n=0}^{\infty} x^n$, where $|x| < 1$ or $-1 < x < 1$

This means that $\dfrac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \ldots$, where $|x| < 1$

Differentiate both sides of $\dfrac{1}{1-x} = 1 + x + x^2 + x^3 + x^4 + \ldots$:

$$\frac{d}{dx}\left[\frac{1}{1-x}\right] = (-1)(1-x)^{-2} \cdot (-1) = \frac{1}{(1-x)^2}$$

$$\frac{d}{dx}\left[1 + x + x^2 + x^3 + x^4 + \ldots\right] = 1 + 2x + 3x^2 + 4x^3 + \ldots$$

Hence, $\dfrac{1}{(1-x)^2} = 1 + 2x + 3x^2 + 4x^3 + \ldots = \sum_{n=0}^{\infty} (n+1)x^n$, where $|x| < 1$

a) The interval of convergence for the power series $\sum_{n=0}^{\infty}(n+1)x^n$ is __?__.

b) The radius of convergence for the power series $\sum_{n=0}^{\infty}(n+1)x^n$ is __?__.

c) Can the function $f(x)=\dfrac{1}{(1-x)^2}$ be expanded into a Taylor series for values near $x_0=0$?

d) Is the function $f(x)=\dfrac{1}{(1-x)^2}$ analytic at at $x_0=0$?

Question 8:

Find a power series representation for the function $f(x)=\dfrac{1}{(1-x)^2}$, centered at $x_0=1$.

a) $f'(1)=$ ___?___

b) Does a power series representation for the function $f(x)=\dfrac{1}{(1-x)^2}$, centered at $x_0=1$, exist?

c) Can the function $f(x)=\dfrac{1}{(1-x)^2}$ be expanded into a Taylor series for values near $x_0=1$?

d) Is the function $f(x)=\dfrac{1}{(1-x)^2}$ analytic at at $x_0=1$?

Question 9:

Find a power series representation for the function $f(x)=\dfrac{1}{x^2+3x+2}$, centered at $x_0=-1$.

a) $f(-1)=$ __?__

b) Does a power series representation for the function $f(x)=\dfrac{1}{x^2+3x+2}$, centered at $x_0=-1$, exist?

c) Can the function $f(x)=\dfrac{1}{x^2+3x+2}$ be expanded into a Taylor series for values near $x_0=-1$?

d) Is the function $f(x)=\dfrac{1}{x^2+3x+2}$ analytic at at $x_0=-1$?

Question 10:

Find a power series representation for the function $f(x)=\dfrac{1}{x^2+3x+2}$, centered at $x_0=-2$.

a) $f(-2)=$ ___?___

b) Does a power series representation for the function $f(x)=\dfrac{1}{x^2+3x+2}$, centered at $x_0=-2$, exist?

c) Can the function $f(x)=\dfrac{1}{x^2+3x+2}$ be expanded into a Taylor series for values near $x_0=-2$?

d) Is the function $f(x)=\dfrac{1}{x^2+3x+2}$ analytic at at $x_0=-2$?

Question 11:

Find a power series representation for $f(x) = x^{3/2}$, centered at $x_0 = 0$.

a) $f'(x) =$ ___?___ and $f''(x) =$ ___?___

b) Does a power series representation for the function $f(x) = x^{3/2}$, centered at $x_0 = 0$, exist?

c) Can the function $f(x) = x^{3/2}$ be expanded into a Taylor series for values near $x_0 = 0$?

d) Is the function $f(x) = x^{3/2}$ analytic at at $x_0 = 0$?